Real Estate Support Triage Agent Report

Course Name:
Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Table of Contents

Problem Statement & Objectives

1. Problem Statement
2. Project Objectives
3. Scope of the Project

Proposed Solution

1. Key features
2. Overall Architecture / Workflow
3. Tools & Technologies Used

Results & Output

1. Screenshots / outputs
2. Reports / dashboards / models
3. Key outcomes

Conclusion

Future Scope & Enhancements

1. Problem Statement & Objective

1.1. Problem Statement

The real estate industry receives a large number of customer communications such as property inquiries, complaints, and support requests. Handling these messages manually can lead to delays, inefficient prioritization, and increased workload for support teams. There is a need for an intelligent system that can automatically analyze incoming messages, identify their urgency and intent, extract important information, and assist in generating quick responses to improve communication efficiency.

1.2. Project Objectives

- Develop a reactive triage agent for real estate communications that can automatically process incoming customer messages such as inquiries, complaints, and service requests.
- Implement real-time text classification to determine the intent and urgency of incoming messages and prioritize them accordingly.
- Apply Named Entity Recognition (NER) to extract critical data points such as property IDs, dates, customer details, and booking information from the messages.
- Integrate Large Language Model (LLM) APIs to generate contextually accurate draft responses for customer queries.
- Improve support workflow efficiency by reducing manual effort and accelerating response and resolution time for real estate support teams.

1.3. Scope of the Project

- Focuses on automated message processing system for real estate domain
- Handles text-based customer interactions
- Provides classification, extraction, and response generation
- Designed as a prototype AI agent system
- Does not include full CRM or enterprise deployment

2. Proposed Solution

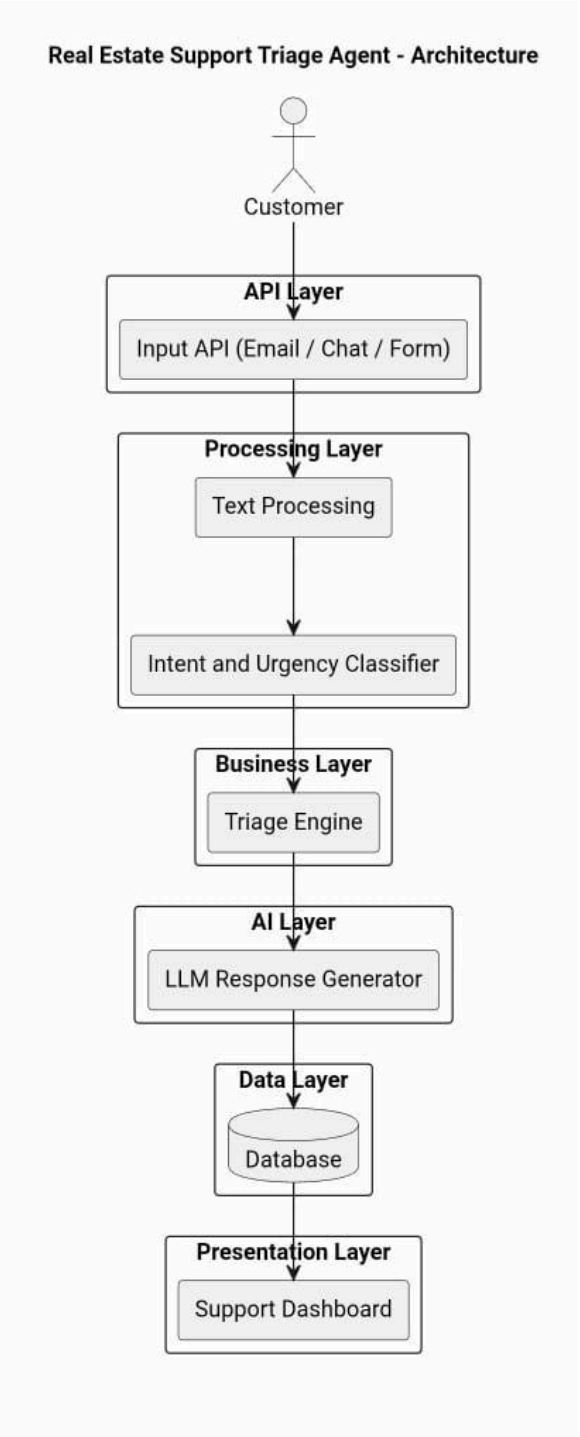
2.1. Key Features

- AI-powered triage agent
- Automatic message classification (intent + urgency)
- Named Entity Recognition (NER)
- Real-time processing of customer queries
- LLM-based response generation
- Prioritization system for handling requests

2.2. Overall Architecture / Workflow

- **User Input via API Layer :** Customers interact through chat, email, or form inputs, which are handled by the API layer.
- **Text Processing with Rule-Based NLP :** Input is processed using basic NLP techniques, including keyword-based NER and rule-based extraction to identify important entities and context.

- **Intent & Triage Decision Making :** The system classifies intent and urgency, and the triage engine decides the appropriate response or action.
- **AI Response & Data Handling :** The LLM generates intelligent responses, data is stored in MongoDB, and results are displayed via the support dashboard.



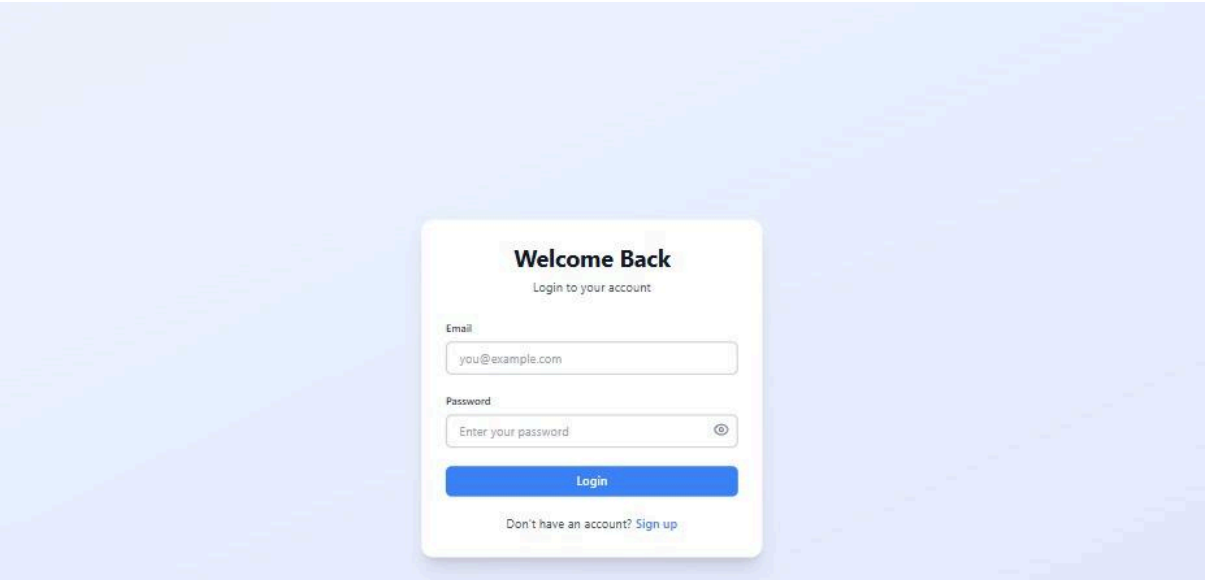
2.3. Tools and Technologies Used

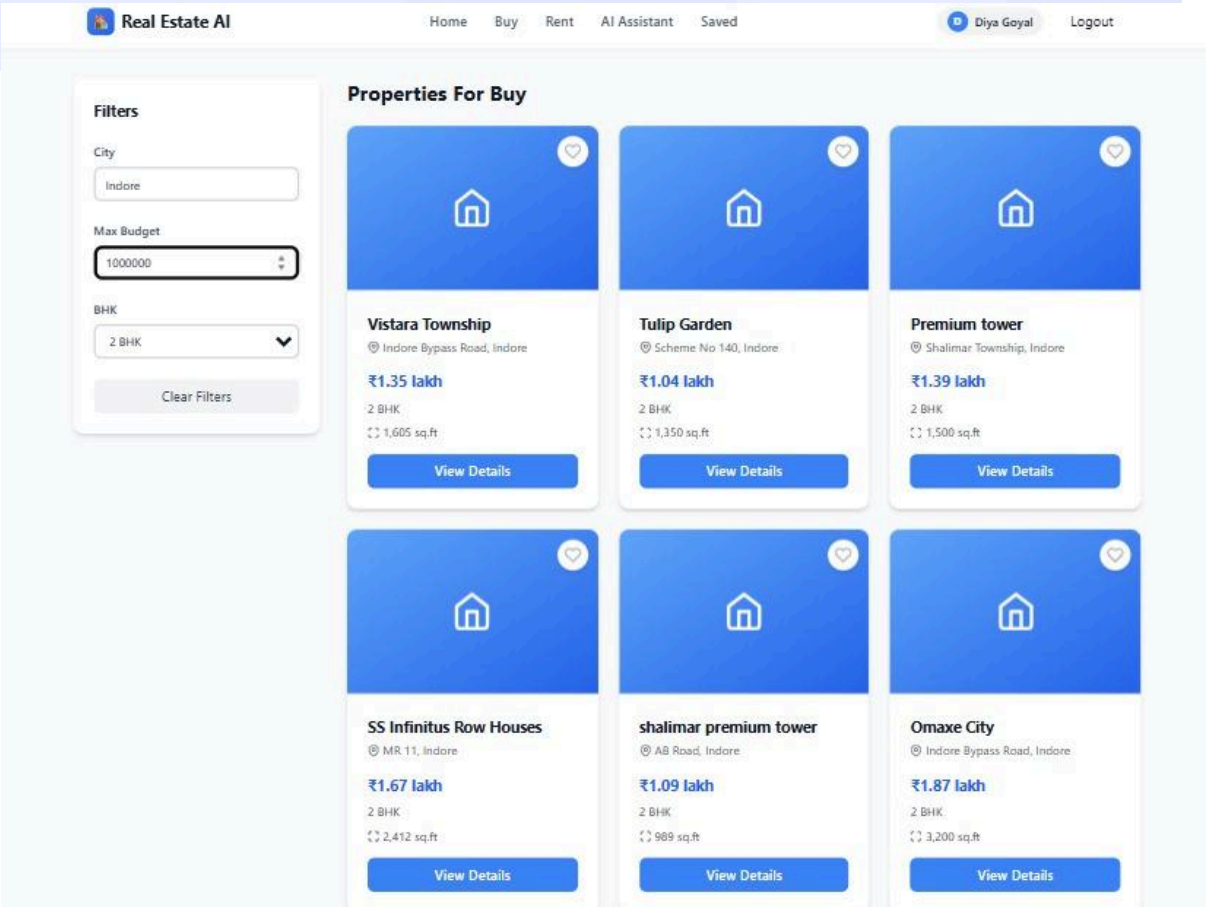
- Programming Languages: Python, JavaScript
- Frontend: React.js
- Backend Framework: FastAPI (Python)
- Backend Technologies: REST APIs, Asynchronous Processing (async/await)
- Database: MongoDB (NoSQL)
- Data Collection: Web Scraping (Scraper-based system)
- AI / LLM Integration: Gemini-2.5-flash
- AI Orchestration: LangChain
- Natural Language Processing: LLM-based NLP (via LangChain)
- Machine Learning: Pre-trained Large Language Models (No custom training)
- Data Format: JSON
- Authentication: JWT (JSON Web Tokens)
- Development Tools: VS Code

3. Results & Output

3.1. Screenshots / System outputs

The system outputs are displayed through an interactive interface, showing the processed results of user inputs. The screenshots include examples of input messages along with their classified intent, detected urgency level, extracted entities (such as location, property type, and budget), and the generated response. These outputs demonstrate the working of the Real Estate Triage Agent in real-time, highlighting its ability to analyze and respond to customer queries efficiently.


Welcome Back
Login to your account
Email
you@example.com
Password
Enter your password
Login
Don't have an account? Sign up


Real Estate AI
Home
Buy
Rent
AI Assistant
Saved
Diya Goyal
Logout
Filters
City
Indore
Max Budget
1000000
BHK
2 BHK
Clear Filters
Properties For Buy
Vistara Township
Indore Bypass Road, Indore
₹1.35 lakh
2 BHK
1,605 sq.ft
View Details
Tulip Garden
Scheme No 140, Indore
₹1.04 lakh
2 BHK
1,350 sq.ft
View Details
Premium tower
Shalimar Township, Indore
₹1.39 lakh
2 BHK
1,500 sq.ft
View Details
SS Infinitus Row Houses
MR 11, Indore
₹1.67 lakh
2 BHK
2,412 sq.ft
View Details
shalimar premium tower
AB Road, Indore
₹1.09 lakh
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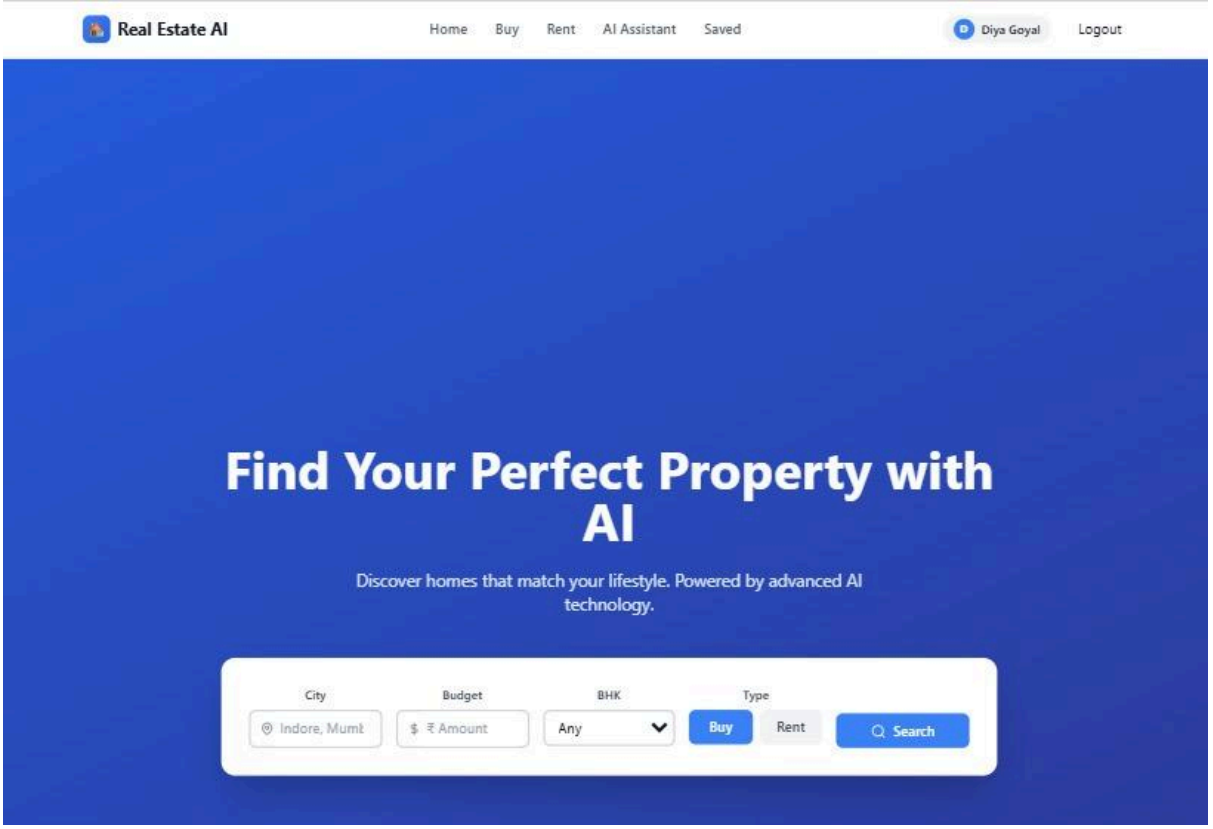
3,200 sq.ft
View Details


Real Estate AI
Home
Buy
Rent
AI Assistant
Saved
Diya Goyal
Logout
Find Your Perfect Property with AI
Discover homes that match your lifestyle. Powered by advanced AI technology.
City
Budget
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Buy
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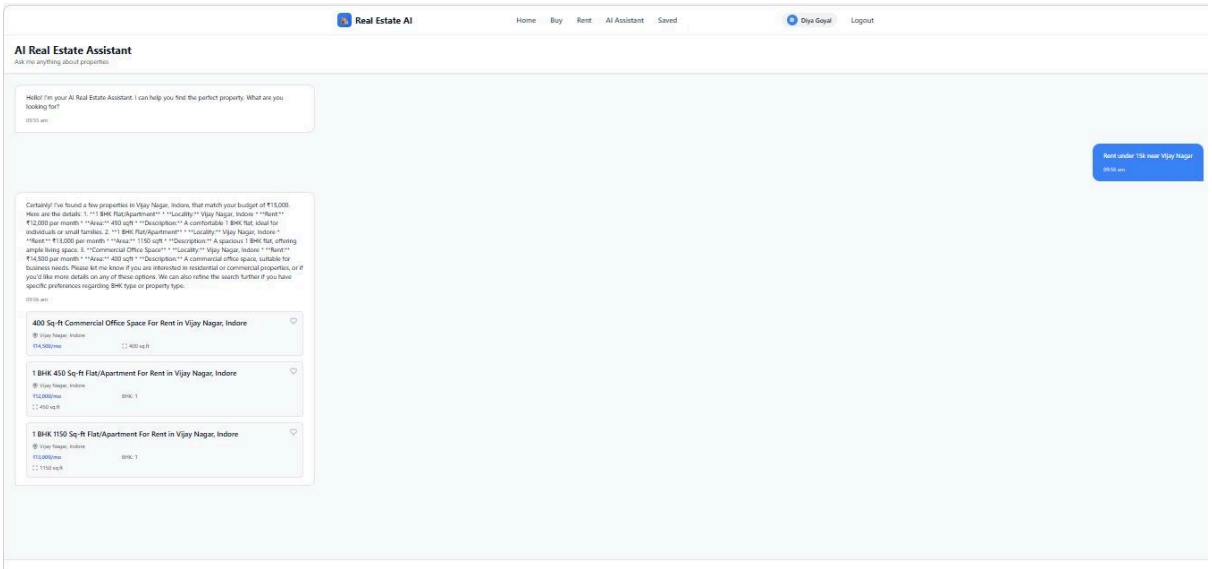
Real Estate AI
Home
Buy
Rent
AI Assistant
Saved
Diya Goyal
Logout
AI Real Estate Assistant
Ask me anything about properties
Hello! I'm your AI Real Estate Assistant. I can help you find the perfect property. What are you looking for?
Rent under 15k near Vijay Nagar
400 Sq-ft Commercial Office Space For Rent in Vijay Nagar, Indore
1 BHK 450 Sq-ft Flat/Apartment For Rent in Vijay Nagar, Indore
1 BHK 1150 Sq-ft Flat/Apartment For Rent in Vijay Nagar, Indore

3.2. Key Outcomes

- **Automated Query Handling:** The system can automatically process and respond to real estate queries with minimal human intervention.
- **Accurate Intent Detection:** Successfully identifies user intent (e.g., buying, renting, inquiry) to provide relevant responses.
- **Efficient Information Extraction:** Implements keyword-based NER and rule-based extraction to capture key details like location, budget, and property type.
- **Intelligent Response Generation:** Uses LLM integration to generate context-aware and human-like responses.
- **Effective Triage Decision System:** The triage engine routes queries based on detected intent for appropriate handling.
- **Scalable and Modular Architecture:** The layered system design supports easy scalability and future enhancements.
- **Efficient Data Management:** MongoDB enables flexible storage and fast retrieval of property and user data.
- **Improved User Experience:** Provides a smooth chat-based interface and support dashboard for interaction.

4. Conclusion

The Real Estate Triage Agent project successfully demonstrates the application of Agentic AI in automating customer communication within the real estate domain. The system is designed to handle incoming user messages efficiently by leveraging Python as the core development language along with advanced AI frameworks such as LangChain and CrewAI. By integrating Large Language Model APIs provided by OpenAI, the system is capable of understanding natural language, classifying user intent, detecting urgency levels, and generating meaningful, context-aware responses.

Additionally, the use of Named Entity Recognition (NER) techniques through tools like spaCy enables the extraction of critical information such as property details, locations, dates, and budgets from unstructured text. This combination of LLM and NER ensures both accurate understanding and structured data processing.

The implementation of this system significantly reduces manual effort by automating repetitive tasks, improves response time through real-time processing, and enhances the overall customer experience by providing quick and relevant replies. The multi-agent architecture further ensures modularity, scalability, and efficient workflow management.

In conclusion, the Real Estate Triage Agent provides a smart, scalable, and efficient solution for handling real estate communications, demonstrating strong potential for real-world deployment and future enhancements such as chatbot integration, voice-based interaction, and CRM connectivity.

5. Future Scope & Enhancements

The Real Estate Triage Agent has strong potential for further improvement and real-world deployment. Several enhancements can be implemented to make the system more advanced, scalable, and user-friendly.

1. **Integration with Messaging Platforms:** The system can be integrated with platforms like WhatsApp, email, and chatbots to automatically handle real-time customer queries across multiple communication channels.
2. **Voice-Based Interaction:** Future versions can support voice input and output, allowing users to interact with the system through speech, making it more accessible and user-friendly.
3. **Advanced Personalization:** The system can be enhanced to provide personalized property recommendations based on user preferences, past interactions, and search history.
4. **CRM Integration:** Integration with Customer Relationship Management (CRM) systems can help store user data, track interactions, and improve customer relationship handling.
5. **Multilingual Support:** Adding support for multiple languages will enable the system to serve a wider audience and improve accessibility in diverse regions.
6. **Improved Accuracy with Fine-Tuning:** The performance of LLM and NER models can be improved by fine-tuning them on real estate-specific datasets for better accuracy and domain understanding.